

Step 3: The information comes back from HDFS into a disseminated H2O frame. The thin arrows in Figure 4 represent the control information. The thick arrows represent the information entity that recurs from HDFS. The distributed H2O frame cluster memory stores the block of data.

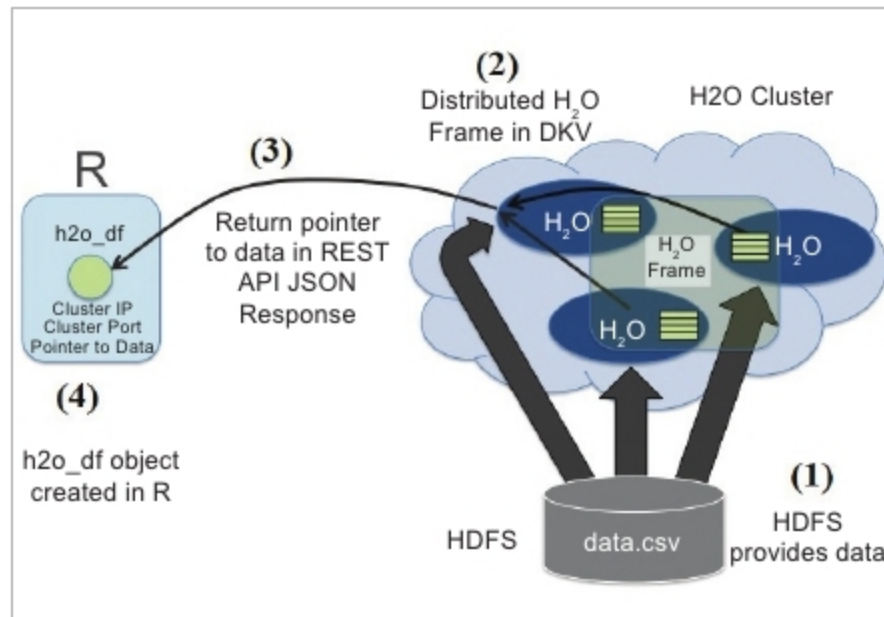


Figure 4: Information comes back from HDFS into a disseminated H2O frame

(ii) *How R scripts call H2O GLM:* Let's use the same H2O application instance created for the R example. It is interesting that you can actually share models created in R and vice-versa. After an H2O application is running, using the R H2O library, you can connect, load data, and train the following GLM:

```
dat <- ...
model <- h2o.glm(y="y",
                 x=c("x1", "x2", "x3"),
                 training_frame=dat)
```

(iii) *How R expressions are sent to H2O for evaluation:* With the S3 object, the class H2O data frame is illustrated by R expressions. The S3 object has an id characteristic which is a reference to the big data object inside H2O.

An introduction to H2O AutoML

H2O AutoML automates the training and tuning of many machine learning process models within a prescribed client enumerated time limit. In the case of Stacked Group (Ensembles), one form is based on all formerly instructed models and the other one is the foremost model of every family – it will be mechanically instructed on a group of individual models to build extremely anticipated ensemble models which, in major instances, will be the best functioning models in the AutoML Leaderboard.

The AutoML interface: This AutoML interface has been designed to create parameters that are manageable so that all that the client requires is to spot its data set, state the time restrictions and identify the response standard.

In both the R and Python API, AutoML utilises the same data related arguments:

`x, y, training_frame, validation_frame`

..where,

- To eliminate predictor columns from data sets, the user will need to specify the `x` argument.
- The response column is represented by the name or index called the `y` argument. This is required.
- The training set is the `training_frame`. This is required.
- To prematurely stop the training process of the independent models in AutoML, run the `validation_frame`, which is used optionally.
- To specify the particular data frame to grade models, the `leaderboard_frame` argument is used, and this frame will not be used for anything besides the leaderboard.
- The user can state the following: `max_runtime_secs`, normal to 600 seconds, to control how extensively the AutoML will run.

AutoML output: In binary grouping problems, the metric is AUC; and in multiple grouping problems, the metric is mean per-class error. In the regression problem, the standard sort measure is Root Mean Square Error (RMSE). The AutoML models were instructed in the leaderboard process, and are classified by a typical metric deployed on the problem type.

NLP with H2O

This segment focuses on a pioneering algorithm — H2O's Word2Vec. This instructs the schematic neural network model to assimilate vector description of words. These vector descriptions are expected to capture the sense of the words.

Supervised learning with text data using Word2Vec:

This process is split up into three phases.

Phase 1: The system will instruct a prototype by using numerical predictor variables. For example, it learns vector representations of words by analysing a large text corpus. Word embedding is nothing but mapping of words to vectors from a high dimensional space (100-1000). Examples of text sources are Google News, Wikipedia, tweets, etc.

Phases 2 and 3: The system will instruct a prototype by using text columns. And the text columns in this information are the product analysis and synopsis.

Embeddings capture meanings of the word:

Semantically similar words are close to each other. H2O can be used to find synonyms and analogies. For example: Man is to woman as king is to ??? (queen).

`Vec(king) - Vec(man) + Vec(woman) = Vec(queen)`

To control our text columns, the system will instruct a Word2Vec prototype to transform text into numeric vectors.

H2O security

H2O contains security highlights proposed for deployment inside a secure data centre (see Table 1).